TASA TKU AEROSPACE

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Regulations of 2026 Orbital Design Contest Preliminary Round

1 Competition Objective

This competition aims to cultivate participants' practical capabilities in **orbital design**, **space mission analysis**, and **orbital maneuver planning**. Participating teams are required to design the maneuver strategy of the "Earth spacecraft" (hereafter referred to as Spacecraft B) in order to intercept the "alien spacecraft" (hereafter referred to as Spacecraft A) within a limited time while considering both fuel consumption and mission efficiency.

2 Mission Setup

- **Spacecraft A:** Subject only to gravitational forces and performs no active maneuvers.
- **Spacecraft B:** Capable of executing multiple impulsive maneuvers ($\Delta \mathbf{v}_k$).
- **Maneuver Constraints:**
  1. Each impulsive maneuver must satisfy the velocity increment constraint:

     $$||\Delta \mathbf{v}_k|| \leq \Delta V_{\text{lim}}$$

     In this competition, the magnitude of each acceleration or deceleration maneuver shall not exceed 1500 m/s, i.e.,

     $$\Delta V_{\text{lim}} = 1500 \text{ m/s}$$

     Additionally, there must be at least 100 seconds between any two maneuvers.
  2. The first maneuver may be executed immediately at the start of the mission.

3 Mission Time Limit

Let the orbital period of Spacecraft A be denoted as T_A. The mission time limit is defined as four times this orbital period (T_{max}). Participants must complete all maneuvers and interception actions within this time window:

$$T_{\text{max}} = 4T_A$$

4 Interception Success Criteria

The relative distance is defined as:

$$\Delta r(t) = ||\mathbf{r}_B(t) - \mathbf{r}_A(t)||$$

If there exists a time t^* satisfying

$$\Delta r(t^*) \leq 5 \text{ km}$$

for some $t \leq T_{\text{max}}$, the interception is considered successful.

- **Interception Time** (t_{int}): The **first** time satisfying the interception condition. That is, if

  $$t^* = \{t_1^*, t_2^*, \cdots\}$$

  then

  $$t_{\text{int}} = t_1^*$$

- **Mission Completion Time** (T_{team}): Defined as either the interception time t_{int} or $4T_A$.

Thus,

$$\begin{aligned} T_{\text{team}} &= \min(t_{\text{int}}, 4T_A) \\ \Delta r_{\text{min}} &= \max\left(\Delta r(T_{\text{team}}), 5\right) \end{aligned}$$

Note: In this preliminary round, only relative position is considered. There is **no restriction on relative velocity** during close approach.

5 Scoring Method

The total velocity increment ΔV_{team} is defined as:

$$\Delta V_{\text{team}} = \sum_{k=1}^{N} \Delta V_k \tag{1}$$

where N is the total number of maneuvers.

The scoring formula considers interception performance, fuel efficiency, and time efficiency:

$$\text{Score} = 50 \exp\left(-\frac{(\Delta r_{\text{min}} - 5)}{100}\right) + \frac{25}{1 + e^{k_t(t_{\text{team}} - C_t)}} + \frac{25}{1 + e^{k_v(\Delta V_{\text{team}} - C_v)}} - \sum_n P_n \tag{2}$$

The parameters (k_t, C_t, k_v, C_v) depend on the orbital distribution conditions and will be announced before each competition.

Please note that any violation of the competition rules during the design process will result in penalties. For example, if a maneuver exceeds the velocity increment limit,

$$\Delta V_k > \Delta V_{\text{lim}}$$

a penalty of 10 points will be deducted for each violating maneuver, i.e.,

$$P_n = 10$$

until the total score reaches zero.

6 Tie-Breaking Rules

If multiple teams obtain the same score, rankings will be determined according to the following priority order:

Priority	Criterion	Decision Rule
1	Minimum Relative Distance $d_{\text{min,team}}$	Smaller distance ranks higher
2	Total Velocity Increment ΔV_{team}	Lower consumption ranks higher
3	Mission Completion Time T_{team}	Shorter time ranks higher
4	Design Theory	Teams with tied scores will give a 5-minute presentation explaining their orbital design methodology.

7 Final Round Qualification

The top 8 teams in the preliminary ranking will advance to the final round. The final round will feature more challenging scenarios and dynamic environmental conditions.

8 Competition Supplementary Rules

- All results are subject to verification by the organizer's official validation program.
- If a submitted result cannot be reproduced, the organizer reserves the right to disqualify the score.
- The organizer reserves the right of final interpretation of all competition rules.
- Any revisions to the competition rules will be announced simultaneously on the official National Space Organization competition website, and the latest revised version shall prevail.